



THE UNITED REPUBLIC OF TANZANIA
MINISTRY OF FINANCE
INSTITUTE OF ACCOUNTANCY ARUSHA



FORM NO. 3

25-09-2026

LOG BOOK

Student Name: ABDULKADIR OMAR

Registration Number: BCSE-01-0036-2024

Programme: Bachelor Of Cyber Security

Name of the Organization: YanaCorp

Department: Information And Communications Technology

On-site Supervisor's Name: Elisha Alex Mkumbo -

DATE	BRIEF DESCRIPTION OF DUTIES PERFORMED
03 Aug, 2026	Finalized the university end-of-semester examinations.
04 Aug, 2026	Travelling day from Arusha to Dar es Salaam. Settled in and prepared for field practical training for the next day.
05 Aug, 2026	Introduction to Git & GitHub Version Control Workflows: 1. Initialized a local Git repository on the Kali Linux terminal and practiced basic staging and committing commands (git status, git add, git commit). 2. Formulated clean branching methodologies by establishing isolated feature branches (login-page). 3. Successfully merged multiple design structures back into the primary production branch (main).



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06 Aug, 2026	Remote Repository Management and Troubleshooting: 1. Created a secure remote repository on GitHub and established remote connections using the command-line interface. 2. Resolved advanced repository migration tasks by using git remote set-url origin to seamlessly redirect a cloned external codebase into a personal repository. 3. Practiced secure CLI authentication standards by configuring and using GitHub Personal Access Tokens (PAT) instead of standard passwords.
07 Aug, 2026	Introduction to Enterprise SOC Tools and Capabilities: 1. Introduced to the primary Security Operations Center (SOC) toolset utilized by the team: Wazuh, Cortex, and TheHive. 2. Conducted foundational research to understand the core functionalities, purposes, and roles each tool plays in threat monitoring and incident response. 3. Studied how these platforms integrate to detect alerts, orchestrate automated analysis, and manage security incidents within a live operational environment.
08 Aug, 2026	Weekend - Off duty
09 Aug, 2026	Weekend - Off duty
10 Aug, 2026	Task/Learning: Installed and configured Windows Server 2019 on Virtual Machine. Set the Administrator password, renamed the server to DC01, configured a static IP address, and learned the fundamentals of Active Directory, Organizational Units (OU), web certificates and Active Directory Certificate Services (AD CS).
11 Aug, 2026	Task/Learning: Installed and configured Ubuntu Server 22.04. Configured a static IP address using Netplan, installed and verified Nginx, installed MariaDB, created a database user named Abdul, and granted the all required privileges. Challenge: Initially faced challenges configuring the static IP, especially identifying the correct network interface, gateway and Netplan configuration. The issue was resolved by checking the network configuration and testing connectivity.
12 Aug, 2026	Task/Learning: Configured and worked with the Active Directory domain yana.asulo. Learned about certificates and PKI, including normal and wildcard certificates, and practiced the concept of using public and private SSH keys for secure authentication. Challenge: Faced challenges understanding the relationship between Active Directory, DNS, certificates and key-based authentication. I learned how certificates establish trust and how SSH keys can be used instead of password-based authentication.

ALL COMMUNICATION TO BE ADDRESSED TO THE RECTOR

Njiro Hill, P.O. Box 2798, Arusha, **Tel:** +255 27 2970232 **Mob:** +255 763 462109 **Telex:** 50009 IAA TZ
Fax: +255 27 2970234 **Email:** iaa@iaa.ac.tz **Website:** www.iaa.ac.tz



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13 Aug, 2026	Task/Learning: Continued working with the Active Directory environment and DNS configuration. Practiced configuring the domain environment and verifying name resolution for yana.asulo. Also continued understanding secure authentication using SSH keys and the importance of protecting private keys. Challenge: Encountered challenges during DNS and network configuration. Troubleshooting was performed by checking IP configuration, DNS settings and connectivity between the systems.
14 Aug, 2026	Task/Learning: Created an Organizational Unit named Lab inside Active Directory and created a Security OU within it. Created two domain users and practiced managing users and access within the domain. Also worked on Windows 11 domain integration and user access. Challenge: The main challenge was ensuring that the Windows 11 machine could communicate correctly with the Domain Controller and resolve the yana.asulo domain. This helped me understand the importance of correct DNS configuration when joining a Windows machine to an Active Directory domain.
15 Aug, 2026	OFF – Weekend
16 Aug, 2026	OFF – Weekend
17 Aug, 2026	Task: Endpoint Hostname Standardization & Wazuh Agent Configuration Standardized employee workstation hostname to match company conventions (YC-Develop-004). Managed the Wazuh service (NET STOP wazuh) and re-authenticated the agent with wazuh.yana.sec under the Computers group. Coordinated with the Infrastructure Supervisor to ensure IP address authorization on the network firewall before restarting the service (NET START wazuh).
18 Aug, 2026	Task: Endpoint Maintenance & Active Directory OU Verification Uninstalled legacy Wazuh agents and clean-installed updated versions across employee Windows workstations, confirming proper enrollment. Verified Active Directory structures, validating the parent Lab OU, nested Security OU, and user account integrity in yana.asulo.
19 Aug, 2026	Task: Burp Suite Proxy Setup & Web Security Fundamentals Integrated Burp Suite Community Edition with Mozilla Firefox via FoxyProxy extension. Analyzed real-time HTTP Request and Response packets using Burp Proxy. Utilized Burp Repeater to modify and re-send HTTP payloads, conducting practical assessment labs on Path Traversal and Broken Access Control vulnerabilities.

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20 Aug, 2026	Task: Web Application Security Labs (Authentication & SSRF) Completed practical modules on Authentication mechanisms, testing session management flaws and credential brute-forcing. Completed advanced hands-on labs (6/6) for Server-Side Request Forgery (SSRF) on PortSwigger Web Security Academy using Burp Repeater to manipulate internal HTTP requests.
21 Aug, 2026	Task: File Upload Vulnerabilities & Remote Code Execution (RCE) Studied mechanisms of unrestricted file upload vulnerabilities on web applications. Practiced bypassing MIME-type validations and file extension controls via Burp Suite. Successfully uploaded custom PHP web shells to execute server-side commands and achieve Remote Code Execution (RCE).
22 Aug, 2026	Weekend / Off
23 Aug, 2026	Weekend / Off
24 Aug, 2026	Task: Metasploit Framework Setup & Service Enumeration. Conducted service enumeration on the Metasploitable sandbox environment using nmap -sV. Configured and launched Metasploit Framework console (msfconsole). Mastered the core 5-step exploitation workflow: search, select module, configure target (RHOSTS/LHOST), select payload, and execute (exploit). Executed practice exploits targeting vulnerable network services (vsftpd 2.3.4 backdoor and Samba usermap_script).
25 Aug, 2026	Public Holiday / Off Official Public Holiday in observance of Maulid ya Mtume Muhammad (S.A.W).
26 Aug, 2026	Task: Advanced Path Traversal Mastery & Authentication Testing Setup. Completed the entire PortSwigger Practitioner Path Traversal module (14/14 labs). Practiced bypass techniques including Absolute Paths, Nested Traversals, Double URL Encoding, and Null Byte Injection. Initiated the Authentication vulnerability module, configuring Burp Intruder with Grep - Extract rules for response error matching and username enumeration.
27 Aug, 2026	Task: Response Timing Enumeration & Brute-Force Protection Bypasses. Completed Practitioner Lab: Username enumeration via response timing, analyzing delays in server response times to identify valid usernames. Analyzed flawed brute-force protection logic (account lockouts and IP rate-limiting weaknesses). Utilized Burp Suite to execute targeted credential brute-forcing while circumventing rate-limiting mechanisms.

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28 Aug, 2026	Task: Authentication Vulnerability Module Completion & Review. Finalized practical labs under the Authentication learning path on PortSwigger Web Security Academy. Practiced multi-stage authentication bypass techniques, password brute-forcing strategies, and session handling controls. Documented attack procedures and remediation practices for securing authentication mechanisms against automated attacks.
29 Aug, 2026	Weekend / Off
30 Aug, 2026	Weekend / Off
31 Aug, 2026	Task: Practitioner Authentication Vulnerabilities & Session Cookie Brute-Forcing. Advanced through PortSwigger Authentication path from lab 21 to 36 of 55 completed tasks. Analyzed stay-logged-in cookie structures, hashing mechanisms (Base64, MD5/SHA-1), and session persistence logic. Completed Lab: Brute-forcing a stay-logged-in cookie using Burp Suite Intruder to hijack target user sessions.
01 Sep, 2026	Task: Server-Side Request Forgery (SSRF) Exploitation & Defense Bypasses. Transitioned to PortSwigger SSRF learning path and completed core defense bypass modules (6 of 6 labs). Analyzed SSRF attack impacts targeting internal network infrastructure (http://127.0.0.1/admin). Practiced circumventing blacklists (127.0.0.1 / localhost bypasses using alternative IP encodings) and whitelist parser defenses via Burp Suite.
02 Sep, 2026	Task: Advanced SSRF Module Completion (Blind SSRF & Hidden Surfaces) Completed advanced SSRF tracks: Blind SSRF vulnerabilities (5/5) and Finding hidden attack surface for SSRF vulnerabilities (4/4). Practiced detecting out-of-band (OAST) DNS/HTTP interactions to confirm Blind SSRF vectors. Exploited hidden attack surfaces via HTTP header manipulation (X-Forwarded-For, Referer) and internal IP scanning. Note/Blocker: Identified limitation requiring Burp Collaborator (Burp Suite Pro license) for out-of-band payload execution.
03 Sep, 2026	Task: MFA Bypasses & Other Authentication Mechanisms (Labs 37–47). Resumed Authentication path, completing labs 37 through 47. Fully completed Vulnerabilities in multi-factor authentication (MFA), testing 2FA logic flaws, missing verification steps, and 2FA code brute-forcing. Completed Vulnerabilities in other authentication mechanisms (15 of 15), analyzing password reset flaws and account recovery logic errors.

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04 Sep, 2026	Task: SQL Injection (SQLi) Module Initiation & Core Concepts. Transitioned to the PortSwigger SQL injection learning path. Studied fundamental SQLi concepts, query structures, and database payload delivery mechanisms via user inputs. Completed the introductory lab: What is SQL injection? (1 of 1), executing basic query manipulation to subvert application logic.
05 Sep, 2026	Weekend / Off
06 Sep, 2026	Weekend / Off
07 Sep, 2026	Conducted practical web application security training focusing on SQL Injection (SQLi) vulnerabilities: Examining the database: Enumerated database vendor details, version numbers, and schema tables to understand backend architecture. Retrieving multiple values within a single column: Applied string concatenation techniques to extract combined data fields into a single output stream. Blind SQL injection: Practiced exploiting blind SQLi flaws using conditional responses and time-delay payload techniques to extract data without direct query output.
08 Sep, 2026	Executed initial OpenVAS/GVM environment setup on Kali Linux. Resolved PostgreSQL database service conflicts and initialized system dependencies to start feed synchronization. Encountered setup errors with scan configuration profiles failing to load in the Web UI.
09 Sep, 2026	Resolved GVM feed synchronization and configuration issues, successfully accessing scan profiles (Full and Fast, Discovery). Configured SSH key-based authentication (asulo97) for target Ubuntu Server (192.168.1.241). Executed initial scan and identified privilege elevation failure (su switch user error).
10 Sep, 2026	Reconfigured target server administrative policies by modifying /etc/sudoers via visudo to enable non-interactive execution (asulo97 ALL=(ALL) NOPASSWD: ALL). Updated GVM Credential Manager settings to map elevation tasks to username root using password authentication.
11 Sep, 2026	Executed a full authenticated vulnerability scan on the target Ubuntu Server. Verified that standard SSH login using the key succeeded, but privilege elevation continued to fail during execution, returning an SSH elevation error in the Host Authentications scan report.

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12 Sep, 2026	OFF
13 Sep, 2026	OFF
14 Sep, 2026	Conducted automated vulnerability scanning on the Ubuntu Server infrastructure using OpenVAS/GVM. Focused on evaluating network services, operating system security baselines, and active protocol configurations. Successfully generated a report identifying 5 security flaws with an overall maximum Severity score of 6.5 (Medium), including Kernel speculative execution flaws (RETbleed/SSB), ICMP timestamp leaks, and weak SSH MAC algorithms. Faced a blocker where insufficient privilege elevation prevented deep authenticated scanning.
15 Sep, 2026	Executed remediation and hardening procedures for the identified Ubuntu Server vulnerabilities. Updated Kernel microcode packages (Intel/AMD), applied firewall rules via iptables to drop ICMP timestamp requests/replies, disabled TCP timestamps via sysctl, and restricted SSH to secure MAC algorithms (hmac-sha2-256-etm, hmac-sha2-512-etm, umac-128-etm). Re-scanned the target and verified 100% resolution of all vulnerabilities, bringing severity down to 0.0 (Clean).
16 Sep, 2026	Performed a comprehensive vulnerability scan and security audit on the Windows Server environment using OpenVAS. Evaluated network-exposed SMB/RDP services, active system processes, and patch baselines. Successfully generated a scan report with a maximum Severity rating of 5.0 (Medium), identifying unpatched secondary services, deprecated TLS 1.0/1.1 support, and weak SMB signing requirements. Experienced a challenge with missing domain admin credentials for authenticated registry scanning.
17 Sep, 2026	Applied remediation steps and patch management for the Windows Server endpoint. Deployed pending critical Windows security updates, disabled legacy SSL/TLS protocols via Registry settings, enforced SMB signing policies, and hardened active Group Policy objects (GPO). Re-scanned the Windows Server and confirmed all Medium-severity vulnerabilities were completely mitigated, resulting in a clean security report.
18 Sep, 2026	Advanced Web Application Penetration Testing practice on PortSwigger Web Security Academy. Focused on the SQL Injection module, completing labs on: Exploiting blind SQL injection by triggering conditional responses to exfiltrate password hashes. Error-based SQL injection techniques to force database error messages containing sensitive data. Exploiting blind SQL injection by triggering time delays (e.g., pg_sleep / WAITFOR DELAY) to verify out-of-band injection flaws.

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19 Sep, 2026	Weekend Off.
20 Sep, 2026	Weekend Off.
21 Sep, 2026	I presented the vulnerabilities identified during the vulnerability assessment of the Ubuntu Server. I explained the vulnerabilities discovered, their potential security impact, and the techniques used to identify them. I also presented the remediation and hardening measures applied to eliminate or reduce the identified vulnerabilities. The supervisor reviewed my work and provided guidance on improving the security of the Ubuntu Server.
22 Sep, 2026	I presented the vulnerabilities identified during the security assessment of the Windows Server and Windows 11 systems. I explained the security weaknesses discovered, their potential impact, and the methods used to remediate them. The supervisor reviewed the implemented security measures and provided additional guidance on Windows system hardening and vulnerability management.
23 Sep, 2026	I was assigned a task to install and configure Wazuh on an Ubuntu environment for security monitoring. During the installation and configuration of Wazuh agents, I encountered several errors while attempting to deploy and enroll agents on the Ubuntu Server and Windows Server nodes. I worked on troubleshooting the installation and agent enrollment issues while following the supervisor's guidance.
24 Sep, 2026	I successfully installed and enrolled the Wazuh agents on the required nodes after troubleshooting the previous configuration and installation errors. The supervisor then assigned me a study task to understand the three major components of the Wazuh platform: Wazuh Dashboard, Wazuh Manager, and Wazuh Indexer. I studied the purpose, role, and interaction of each component within the Wazuh security monitoring architecture.
25 Sep, 2026	I continued with practical learning on Wazuh, focusing on security monitoring and understanding how the platform collects, processes, and presents security-related information from monitored endpoints. I explored the monitoring capabilities of Wazuh and how security events and alerts can be analyzed through the Wazuh Dashboard. I also continued learning how the Wazuh Manager and Wazuh Indexer support the monitoring and analysis process.

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26 Sep, 2026	Weekend-Off
27 Sep, 2026	Weekend-Off

PART J: STUDENT STATEMENT

Student's signature: _____

Date: _____

On-site supervisor's signature: _____

Date: _____

(OFFICIAL STAMP)